

PSP database limitations

Context

PSP monolith database is holding objects in 3 tablespaces. These are PSP_DATA01, PSP_IDX01, PSP_LOB01. Out of these PSP_DATA01 is reaching 16 TB (maximum size allowed by RDS on bigfile tablespace). One can read more about this [here](#).

Scope

Scope for this is limited to Monolith Primary and DR databases.

Current status

Tablespace in Monolith database:

Tablespace	Total Size(Mb)	Total Free(MB)	% Free	Max Free(Kb)	No Of Ext.
PSP_DATA01	15,164,726	722,188	4.76	2,097,152	429
PSP_IDX01	9,013,032	429,245	4.76	2,097,152	215
PSP_LOB01	4,632,535	220,645	4.76	2,097,152	188

Database Objects exists in tablespaces

OWNER	SEGMENT_TYPE	TABLESPACE_NAME	COUNT(*)
PSPADM	TABLE	PSP_LOB01	1
PSPADM	LOBINDEX	PSP_LOB01	2
PSPADM	LOBSEGMENT	PSP_LOB01	2
PSPADM	TABLE	APPPBUGFIX_DATA	2
PSPADM	INDEX	PSP_LOB01	11
PSPADM	LOBINDEX	PSP_DATA01	22
PSPADM	LOBSEGMENT	PSP_DATA01	22
PSPADM	INDEX	PSP_IDX01	305
PSPADM	TABLE	PSP_DATA01	360
PSPADM	TABLE PARTITION	PSP_LOB01	385
PSPADM	INDEX	PSP_DATA01	453
PSPADM	TABLE PARTITION	PSP_DATA01	1343
PSPADM	INDEX PARTITION	PSP_IDX01	2668

Indexes which are in PSP_DATA01

OWNER	SEGMENT_TYPE	TABLESPACE_NAME	COUNT(*)	SUM(BYTES)/1024/1024/1024
PSPADM	LOBINDEX	PSP_DATA01	22	0.012
PSPADM	INDEX	PSP_DATA01	453	5126

Database Growth

Database growth is ~260 GB per month (~3 TB per year). Section "RDS OS Disk free Space" <https://intuit.wavefront.com/u/7lgwSwlMIs?t=intuit>.

Approaches

Approach	Pros	Cons	Remarks
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1	Alter index move	• Online operation	• db monitoring while it runs	
2	Object redefinition	• online object reorg/redefinition	• Need to be check for table objects once index move done.	
3	parallel setup	• Can be done on parallel setup	• Need more time and efforts • Needs cutover to new setup	

Proposed solution

- Create below bigfile tablespaces
 - PSP_DATA02 - this will hold future table objects.
 - PSP_IDX02 - this will hold all current index objects (by moving it online) and future index objects.
- Modify table attributes for all partitioned tables, so that future partitions will create in different tablespace.
- Try moving indexes online which are on PSP_DATA01 to different index tablespace.
- Code : [Git](#)

Design Principle

- No data loss
- No data corruption
- No downtime
- No Replication Issue to DR site

Status After Index Move

Learnings from Non-Prod (SYS)

Below indexes failed while rebuild. Based on this, we have excluded these indexes in Prod (DR as well as Primary).

```

alter index PSPADM.PSP_EMPLOYEE_TAXIDENC_I1 rebuild tablespace PSP_IDX02 online
*
ERROR at line 1:
ORA-00604: error occurred at recursive SQL level 1
ORA-01450: maximum key length (3215) exceeded
--
alter index PSPADM.PSP_BANK_ACCOUNT_ENC_I1 rebuild tablespace PSP_IDX02 online
*
ERROR at line 1:
ORA-00604: error occurred at recursive SQL level 1
ORA-01450: maximum key length (3215) exceeded
--
alter index PSPADM.PSP_EMP_COMP_FK_SRC_EMP_ID rebuild tablespace PSP_IDX02 online
*
ERROR at line 1:
ORA-08108: may not build or rebuild this type of index online
--
alter index PSPADM.PSP_ENMT_FEDTAXIDENC_I1 rebuild tablespace PSP_IDX02 online
*
ERROR at line 1:
ORA-00604: error occurred at recursive SQL level 1
ORA-01450: maximum key length (3215) exceeded
--
alter index PSPADM.PSP_COMPANY_FEDTAXIDENC_I1 rebuild tablespace PSP_IDX02 online
*
ERROR at line 1:
ORA-00604: error occurred at recursive SQL level 1
ORA-01450: maximum key length (3215) exceeded
--
alter index PSPADM.PSP_FRAUD_ACCOUNT_ENC_I1 rebuild tablespace PSP_IDX02 online
*
ERROR at line 1:
ORA-00604: error occurred at recursive SQL level 1
ORA-01450: maximum key length (3215) exceeded
--
alter index PSPADM.PSP_COMPANY_CONSENT_I1 rebuild tablespace PSP_IDX02 online
*
ERROR at line 1:
ORA-00604: error occurred at recursive SQL level 1
ORA-01450: maximum key length (3215) exceeded

```

TABLE_NAME	INDEX_NAME	IND_SIZE_GB	COLUMN_NAME	CHAR_LENGTH
PSP_EMPLOYEE	PSP_EMPLOYEE_TAXIDENC_I1	13.6	TAX_ID_ENC	4000
PSP_BANK_ACCOUNT	PSP_BANK_ACCOUNT_ENC_I1	7.9	ACCOUNT_NUMBER_ENC	4000
PSP_ENTITLEMENT_UNIT	PSP_ENMT_FEDTAXIDENC_I1	1.5	FED_TAX_ID_ENC	4000
PSP_COMPANY	PSP_COMPANY_FEDTAXIDENC_I1	1.4	FED_TAX_ID_ENC	4000
PSP_FRAUD_BANK_ACCOUNT	PSP_FRAUD_ACCOUNT_ENC_I1	0.125	ACCOUNT_NUMBER_ENC	4000
PSP_AGENCY	PSP_AGENCY_AGENCYIDENC_I1	0.000061	AGENCY_ID_ENC	4000

DR Database

Tablespace	Total Size(Mb)	Total Free(MB)	% Free	Max Free(Kb)	No Of Ext.
PSP_LOB01	4,688,308	223,228	4.76	2,097,152	161
PSP_IDX01	9,341,840	444,813	4.76	2,097,152	228
PSP_IDX02	4,415,517	210,263	4.76	2,097,152	142
PSP_DATA01	15,434,199	6,063,698	39.29	2,097,152	28,196
PSP_DATA02	1,048,576	1,046,435	99.80	2,097,152	512

Primary Database

Tablespace	Total Size(Mb)	Total Free(MB)	% Free	Max Free(Kb)	No Of Ext.
PSP_LOB01	4,691,966	223,427	4.76	2,097,152	192
PSP_IDX02	4,357,950	207,521	4.76	2,097,152	135
PSP_IDX01	9,110,798	433,904	4.76	2,097,152	221
PSP_DATA01	15,364,140	5,991,865	39.00	2,097,152	28,399
PSP_DATA02	1,048,576	1,046,435	99.80	2,097,152	512